BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Trace Tables and Algorithm Refinement

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Name one useful item to record when correcting a failed test.

2. Which access rights should a user have?

A) Full administrator rights
B) All other users' rights
C) No login checks
D) Only those they need

3. "A printer driver stores only the user's website passwords." Correct this. Explain why.

4. A program reads names from a file and saves results. Correct both claims. Explain why. (a) readLine() writes a line. (b) writeLine(text) only shows text on screen.

Today's objectives

Trace, test and refine algorithms. Apply this to another software scenario.

Key vocabulary - Read, cover, recall

Trace table
Manually records changes to variables and printed outputs as code executes.

Accumulator
A variable updated repeatedly to store a running total.

Expected output
The result that should occur according to the problem requirements.

Refinement
Use discrepancies in the trace to revise instructions and retest.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Trace table	Manually records changes to variables and printed outputs as code executes. Consider this in a points tally.
Accumulator	A variable updated repeatedly to store a running total. Consider this in a shop receipt sum.
Expected output	The result that should occur according to the problem requirements. Consider this in a school class score.
Refinement	Use discrepancies in the trace to revise instructions and retest. Consider this in a three-item total.

I Do - One worked example

A points tally needs trace table and accumulator. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Trace table [1]; Manually records changes to variables and printed outputs as code executes. [1].

3. Accumulator [1]; A variable updated repeatedly to store a running total. [1].

4. Give a specific point and a linked reason for each feature.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a shop receipt sum, explain the purpose of trace table. [2 marks]

2. A school class score needs accumulator. Explain its role. [2 marks]

3. A three-item total uses expected output and refinement. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the total printed by this loop.

```
total = 0
for i = 1 to 3
total = total + i
next i
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO concepts relevant to this topic. [2 marks]

Q3. Explain how accumulator and expected output help in a sports count. [4 marks]

Q4. Explain the roles of trace table, expected output and refinement in a fundraising sum. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A trace table records changes to variable values and outputs as instructions execute. ☐ T ☐ F
2. Manually records changes to variables and printed outputs as code executes ☐ T ☐ F
3. A variable updated repeatedly to store a running total ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 The input, expected output, actual output or identified fault, with the correction. ☐

Q2 D - Only those they need ☐

Q3 It enables communication with and control of a particular peripheral. ☐

Q4 (a) readLine() reads the next line and moves forward. (b) writeLine(text) writes a line to an open file. Open, read or write, then close. Screen output does not save a file. ☐

Code First - Model

```
total = 0
for i = 1 to 3
total = total + i
next i
print(total)
```

Output: 6

We Do - Check the model

1. Trace table [1]; Manually records changes to variables and printed outputs as code executes. [1]. ☐

2. Accumulator [1]; A variable updated repeatedly to store a running total. [1]. ☐

3. Expected output [1]; The result that should occur according to the problem requirements. [1]. Refinement [1]; Use discrepancies in the trace to revise instructions and retest. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Trace table [1]; Expected output [1]. ☐

Q3 Accumulator [1]; A variable updated repeatedly to store a running total. [1]. Expected output [1]; The result that should occur according to the problem requirements. [1]. ☐

Q4 Trace table [1]; Manually records changes to variables and printed outputs as code executes. [1]. Expected output [1]; The result that should occur according to the problem requirements. [1]. Refinement [1]; Use discrepancies in the trace to revise instructions and retest. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: